

Multicalibration Is Necessary for Unbiased Model-Based Prevalence Estimation

Significance

Researchers increasingly use AI systems—including large language models—as measurement devices to estimate how common a phenomenon is in a population. A critical but overlooked problem arises when the target population differs from the one on which the device was validated: standard calibration methods produce biased prevalence estimates. We show that multicalibration—requiring a device to be accurate not just on average but across all relevant subgroups—is necessary and sufficient for unbiased prevalence estimation under arbitrary covariate shift. Our result, confirmed by simulation and empirical application to U.S. employment data, implies that the rapidly growing body of research using AI for measurement is vulnerable to systematic bias unless multicalibration is enforced.

Abstract

Estimating the prevalence of a category in a population using imperfect measurement devices—diagnostic tests, classifiers, or large language models—is fundamental to science, public health, and platform governance. Standard approaches correct for known device error rates, but assume these rates remain stable across populations. We show this assumption fails under covariate shift: when the composition of the target population differs from the calibration population, globally calibrated devices produce biased prevalence estimates. We prove that multicalibration—calibration conditional on all relevant subgroups—is both necessary and sufficient for unbiased prevalence estimation under arbitrary covariate shift, connecting recent theory from algorithmic fairness to a longstanding problem in measurement. A simulation study demonstrates that standard methods (Classify & Count, Rogan-Gladen adjustment, isotonic regression) exhibit bias growing with the magnitude of distribution shift, while a multicalibrated estimator maintains near-zero bias throughout. In an empirical application using the American Community Survey, we estimate employment prevalence under synthetic age distribution shifts across 14 U.S. states. Standard methods show bias of 2–19 percentage points; a multicalibrated predictor reduces bias to under

1.2 percentage points in all scenarios. These results have immediate implications for the expanding use of AI systems as zero-shot measurement devices, where extrapolation to new populations is the primary purpose.

Introduction

Researchers in nearly every field frequently aim to measure the prevalence of a category within a target population. This includes use of diagnostic tests to measure disease prevalence, tracking prevalence of phenomena on online platforms, and quantifying document frequencies in a corpus. Recently, large language models (LLMs) have dramatically expanded the ease and reduced the cost of this kind of measurement. Researchers now routinely deploy LLMs as zero-shot measurement devices to estimate the prevalence of phenomena that previously required expensive manual annotation.

Applications span the social sciences, where LLMs code democracy indicators across countries (Weidmann et al. 2025), classify protest events in news corpora (Overös et al. 2024), and categorize open-ended survey responses at near-human accuracy (Mellon et al. 2024; Gilardi et al. 2023); clinical medicine, where they extract diagnostic attributes from pathology reports (Sushil et al. 2024) and identify goals-of-care discussions in clinical notes (Sibley et al. 2025); and the digital humanities, where they annotate art forms in auction records (Tojima and Yoshida 2025) and convert qualitative text into quantitative variables across multiple languages (Karjus et al. 2025). LLMs are also being advertised for content moderation¹ and LLM “judges” are used to assess the quality and safety of AI systems (Zheng et al. 2023; Yuan et al. 2024). The central promise of these model-based measurements is extrapolation: the ability to estimate prevalence in new populations, time periods, or subgroups where manual annotation is infeasible due to scale or too expensive using traditional human annotation. While model-based quantification is more affordable, it typically comes with the risk of false positive and false negative classification errors. However, commonly used remedies for these errors fail when the target population differs from the population on which the device’s error properties were assessed.

Current quantification methods, such as “Classify & Count” or “Adjusted Count” (Rogan-Gladen), offer adjustments that attempt to correct for imperfect devices (González et al. 2017). However, these methods rely on a fragile assumption: that the device’s error properties (sensitivity, specificity, or calibration) remain static across populations. In reality, a model’s performance often varies drastically across subgroups. When the composition of the target population differs from the source population, these subgroup-specific errors no longer cancel out, leading to biased prevalence estimates.

We propose multicalibration as the necessary and sufficient condition for unbiased prevalence estimation under arbitrary covariate shift. Building on recent

¹<https://openai.com/index/using-gpt-4-for-content-moderation/>

theoretical advances in algorithmic fairness and domain adaptation (Kim et al. 2022), we leverage the fact that a multicalibrated classifier achieves ‘Universal Adaptability’ to solve the prevalence estimation problem. We show that this theoretical guarantee allows for unbiased estimates on any downstream target population without the need for target-specific re-weighting or retraining. This robustness is critical for nearly all practical applications of quantification, including applying a model to previously unseen datasets, tracking changes in prevalence over time, or analyzing subsets of the training population. In contrast, we show that simple calibration or any adjustment-oriented quantification technique is insufficient for unbiased estimation except in trivial and unrealistic settings.

We review existing approaches to the quantification problem, connect the quantification task to the broader theory of domain adaptation via multicalibration, demonstrate the need for multicalibration to achieve unbiased estimates of prevalence in out-of-domain settings, and apply those insights to a few real-world datasets.

Results

Standard calibration methods fail under covariate shift

Consider a binary outcome $Y \in \{0, 1\}$, features X , and a device $h(X) \in [0, 1]$ producing probabilistic predictions. The goal is to estimate the population prevalence $\pi = P(Y = 1)$ in a target population using only the device’s outputs. The key assumption concerns which aspect of the data-generating process changes across populations. Three types of distributional change are commonly distinguished:

- **Covariate shift:** The marginal distribution of features $P(X)$ changes, but the conditional $P(Y|X)$ remains stable. This is natural when features causally influence outcomes ($X \rightarrow Y$): applying a device to a new population changes the mix of features but not the underlying feature-outcome relationships. Most measurement applications fit this pattern—demographics influence employment, text content determines topic relevance, patient characteristics drive diagnoses.
- **Label shift** (prior probability shift): The prevalence $P(Y)$ changes, and the conditional $P(X|Y)$ remains stable. This is natural when outcomes generate features ($Y \rightarrow X$), as in some diagnostic settings where a disease produces symptoms. However, for most measurement tasks where the goal is to classify based on observed features, the $X \rightarrow Y$ direction is more appropriate.
- **Concept drift:** The conditional $P(Y|X)$ itself changes—for example, the meaning of words evolves over time, or a disease mutates to present

differently. No purely statistical post-processing can correct for concept drift; new labeled data from the target distribution is required.

We focus on covariate shift, which is the dominant concern in the measurement settings motivating this paper (see Limitations for further discussion of concept drift).

A device is *globally calibrated* if $\mathbb{E}[Y \mid h(X) = p] = p$ for all prediction values p . Under global calibration, $\mathbb{E}[h(X)] = \mathbb{E}[Y] = \pi$, so the sample mean of predictions is an unbiased prevalence estimate. However, global calibration is a property of a *specific* population. A device calibrated on one population need not be calibrated on another, even under covariate shift with stable $P(Y|X)$.

The failure mechanism is straightforward. Suppose the population consists of subgroups indexed by g , with sizes w_g and group-specific calibration errors $\epsilon_g = \mathbb{E}[h(X) - Y \mid G = g]$. Global calibration requires only that these errors cancel on average: $\sum_g w_g \epsilon_g = 0$. Under covariate shift, the weights change to w_g^* , and the bias in the prevalence estimate becomes $\sum_g w_g^* \epsilon_g$, which is generally nonzero unless every $\epsilon_g = 0$. Crucially, a device can be perfectly globally calibrated—with errors that precisely cancel in the training population—while being substantially miscalibrated on every individual subgroup.

The quantification literature has proposed several methods to correct for device errors (González et al. 2017). Each is vulnerable to the failure mechanism described above. *Classify & Count* binarizes predictions at a threshold τ and reports the positive fraction; the threshold is chosen to match true prevalence on calibration data, but this correspondence breaks under shift. The *Adjusted Count* (Rogan-Gladen) estimator $\hat{\pi} = (\hat{p} - \text{FPR}) / (\text{TPR} - \text{FPR})$ uses sensitivity and specificity from calibration data (Rogan and Gladen 1978). These error rates are weighted averages of group-specific rates; when group proportions shift, the learned rates no longer match the target population’s. The ratio estimator further amplifies errors when $\text{TPR} - \text{FPR}$ is small. The same argument applies to calibration approaches using stable error rates such as the Calibrate-Extrapolate framework (Wu and Resnick 2024a, 2024b). *Isotonic regression* and other global calibration methods learn a single mapping from scores to probabilities, which reflects the calibration population’s composition and need not hold elsewhere.

Multicalibration guarantees unbiased prevalence estimation

The analysis above shows that all standard methods produce biased prevalence estimates under covariate shift, with bias $\sum_g w_g^* \epsilon_g$ that is generally nonzero unless every $\epsilon_g = 0$. This condition—requiring calibration errors to vanish within every subgroup, not just on average—is precisely *multicalibration*.

A predictor $f(X)$ is α -*multicalibrated* with respect to a collection of subgroups $\mathcal{G}$ if, for every group $G \in \mathcal{G}$ and every prediction value v :

$$|\mathbb{E}[Y \mid f(X) = v, X \in G] - v| \leq \alpha,$$

whenever the conditioning event has sufficient probability mass (Hébert-Johnson et al. 2018). This strengthens global calibration by requiring accuracy not just on average, but within each subgroup at each score level.

The connection to prevalence estimation is direct. If f is multicalibrated with respect to $\mathcal{G}$, then $\mathbb{E}[f(X) \mid X \in G] \approx \mathbb{E}[Y \mid X \in G]$ for every $G \in \mathcal{G}$. Under covariate shift—where only $P(X)$ changes while $P(Y|X)$ remains stable—the law of iterated expectations gives:

$$\mathbb{E}^*[f(X)] = \sum_g w_g^* \cdot \mathbb{E}[f(X) \mid G = g] \approx \sum_g w_g^* \cdot \mathbb{E}[Y \mid G = g] = \mathbb{E}^*[Y] = \pi^*.$$

Because multicalibration ensures predictions are correct *within each subgroup*, the average of predictions tracks the true prevalence under *any* reweighting of those subgroups. This is the “Universal Adaptability” result of Kim et al. (2022): a multicalibrated predictor competes with propensity scoring for arbitrary target distributions, without requiring knowledge of the shift.

Returning to the framework from the previous section: global calibration only ensures $\sum_g w_g \epsilon_g = 0$ for the training weights w_g . Multicalibration ensures $\epsilon_g \approx 0$ for each subgroup individually, making the estimate robust to arbitrary changes in w_g . This is also a *necessary* condition: if any $\epsilon_g \neq 0$, there exists a target distribution (one that overweights group g) under which the prevalence estimate is biased.

Practical post-hoc multicalibration algorithms can be applied to any base predictor—including LLMs and black-box classifiers—by iteratively correcting predictions on subgroups where they are miscalibrated (Hébert-Johnson et al. 2018; Gopalan et al. 2022). A key challenge is that these algorithms typically require the collection of subgroups $\mathcal{G}$ to be specified explicitly, which practitioners may struggle to do a priori. More recent methods address this by taking a set of segment features as input and automatically discovering miscalibrated regions in the feature space, removing the need for manual group definitions (Tax et al. 2025)².

Simulation: standard methods fail, multicalibration succeeds

We illustrate the theoretical results above with a controlled simulation. We generate data with a binary covariate $X \in \{0, 1\}$ and binary outcome Y , where

²An open-source implementation is available at <https://mcgrad.dev>.

$P(Y = 1|X = 1) = 0.85$ and $P(Y = 1|X = 0) = 0.15$. A classifier produces systematically biased predictions: 10% underestimation when $X = 0$ and 10% overestimation when $X = 1$. All calibration parameters are learned on a balanced training distribution ($P(X = 0) = 0.5$). We then evaluate prevalence estimates on test distributions where $P(X = 0)$ ranges from 0.01 to 0.99, repeating 50 times (details in Materials and Methods).

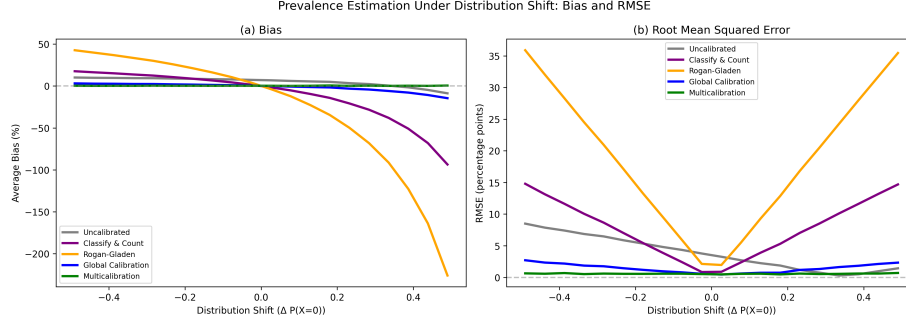


Figure 1: Figure 1

Figure 1: Prevalence estimation under covariate shift for five methods across 50 simulation runs. (a) Average bias (%). (b) Root mean squared error (RMSE). Only multicalibration maintains near-zero bias and RMSE across all levels of distribution shift.

Figure 1 shows the results. At the training distribution (center), all methods except the uncalibrated baseline produce approximately unbiased estimates. As the distribution shifts, the methods diverge. Classify & Count and Rogan-Gladen show rapidly growing bias, exceeding $\pm 40\%$ at extreme shifts. The Rogan-Gladen estimator is particularly unstable: its ratio structure amplifies estimation errors, producing bias that exceeds the other methods by an order of magnitude. Global calibration shows more moderate but still substantial bias (up to $\pm 15\%$). The RMSE panel (Figure 1b) shows that these bias differences dominate total estimation error: the RMSE curves closely track the absolute bias, confirming that variance is small relative to bias for all methods at this sample size. The multicalibrated estimator maintains near-zero bias and the lowest RMSE across the entire range of distribution shifts.

Empirical application: employment prevalence under age distribution shift

To illustrate how distribution shift can lead to strongly biased measurements under distribution shift, we analyze data from the American Community Survey (ACS), a large-scale annual survey conducted by the U.S. Census Bureau. The concept to be measured is the rate of employment. The measurement device is

a predictive model of binary employment status, with 16 sociodemographic features including age, education, marital status, disability status, and citizenship.

Setup. We train a logistic regression classifier on data from eight U.S. states (TX, MI, PA, OH, IL, GA, NC, VA) across 2016–2018, totaling approximately 1.5 million training observations. The remaining in-distribution data is split into a calibration set ($n \approx 644,000$) and a test set ($n \approx 920,000$). Six additional states (CA, NY, FL, WA, AZ, CO) are held out for out-of-distribution (OOD) evaluation. We fit two post-hoc calibration methods on the calibration set: isotonic regression (global calibration) and MCGrad, a multicalibration algorithm that enforces calibration conditional on subgroups defined by both categorical and numerical features. We compare five prevalence estimation methods: raw (uncalibrated) scores, Classify & Count with a prevalence-matched threshold, Rogan-Gladen adjustment, isotonic regression, and MCGrad.

Age distribution shift. Employment rates vary dramatically by age: approximately 47% for ages 16–24, 76% for ages 25–54, 61% for ages 55–64, and 17% for ages 65+. This makes age an ideal dimension along which to construct meaningful distribution shifts. We create synthetic target populations by re-sampling test data with shifted age distributions: young-skewed (oversampling ages 16–30), old-skewed (oversampling ages 60+), and bimodal (oversampling both tails). The resulting populations have true employment rates ranging from 12.8% to 46.0%. All calibration parameters are estimated once on the original calibration set and held fixed across scenarios.

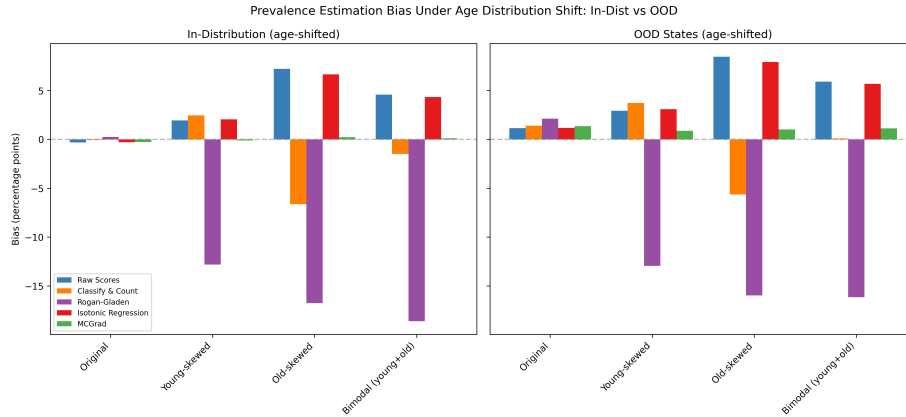


Figure 2: Figure 2

Figure 2: Prevalence estimation bias (percentage points) under synthetic age distribution shift, for in-distribution data (left) and out-of-distribution states (right).

Setting	Age Dist.	True Prev.	Raw Scores	CC	Rogan- Gladen	Isotonic	MCGrad
In-Dist	Original	46.0%	-0.31	-0.07	+0.26	-0.30	-0.27
In-Dist	Young-skewed	12.8%	+1.93	+2.47	-12.82	+2.04	-0.11
In-Dist	Old-skewed	16.8%	+7.23	-6.62	-16.77	+6.65	+0.22
In-Dist	Bimodal	21.1%	+4.57	-1.50	-18.62	+4.33	+0.12
OOD	Original	45.1%	+1.15	+1.40	+2.12	+1.17	+1.35
OOD	Young-skewed	13.0%	+2.93	+3.73	-12.96	+3.08	+0.88
OOD	Old-skewed	16.0%	+8.47	-5.64	-15.97	+7.91	+1.01
OOD	Bimodal	20.8%	+5.91	+0.09	-16.14	+5.69	+1.13

Table 1: Prevalence estimation bias in percentage points (pp) under synthetic age distribution shift.

Results. With no age shift, all methods produce approximately unbiased estimates (Table 1). Under shift, the methods diverge sharply.

Rogan-Gladen exhibits catastrophic failure, with bias of -12.8 to -18.6 percentage points in the in-distribution setting—effectively estimating zero or negative prevalence when the true rate is 13–21%. This occurs because TPR and FPR, estimated on the original age distribution, become invalid when the age mix shifts. Isotonic regression and raw scores show substantial bias of 2–8 percentage points: their global calibration curve, accurate on average in the training distribution, no longer reflects the true relationship when particular age groups are over- or under-represented. Classify & Count shows bias of up to 6.6 percentage points as the learned threshold loses its validity.

MCGrad produces near-zero bias across all in-distribution age-shift scenarios (≤ 0.27 pp). Because MCGrad is multicalibrated with respect to age and other features, its predictions are correct *conditional on* these covariates. When the age distribution shifts, the subgroup-level predictions remain valid, and their average tracks the true prevalence. Bootstrap resampling (200 iterations per scenario) confirms that MCGrad also achieves the lowest RMSE, indicating that the bias reduction does not come at the cost of increased variance.

In the OOD setting—applying the model to states never seen during training *and* shifting the age distribution—MCGrad’s advantage persists with modestly larger bias (0.88–1.35pp), reflecting the additional geographic shift along an axis not included in the calibration subgroups. Even so, MCGrad’s bias remains 3–8 $\times$ smaller than the next best method.

Discussion

We have shown that multicalibration—calibration conditional on subgroups, not just on average—is the key property for accurate model-based prevalence estimation under population shift. Standard calibration methods can produce severely biased estimates when the composition of the target population differs from the calibration population. Multicalibration resolves this by ensuring accuracy at the subgroup level, making prevalence estimates robust to arbitrary reweightings.

The practical implications are immediate. The growing use of LLMs as zero-shot measurement devices rests on the assumption that models validated on one population can be reliably applied to others. Our analysis shows this assumption is safe only when the model is multicalibrated with respect to features that may shift across populations. Without this, prevalence estimates in new populations—the primary use case for LLM-based measurement—are unreliable. Researchers using AI-based measurement should treat multicalibration not as an optional fairness criterion, but as a methodological requirement for valid inference.

This result also clarifies why certain widely-used practices are inadequate. The common approach of reporting a model’s overall accuracy, F1, or AUC on a held-out set provides no information about calibration, let alone multicalibration. A model with 95% accuracy can still produce heavily biased prevalence estimates if its errors are unevenly distributed across subgroups. Similarly, reporting a single calibration curve—as is standard in the quantification literature (González et al. 2017)—certifies only global calibration, which we have shown is insufficient under shift. Even comprehensive methodology textbooks for text-as-data research devote extensive attention to classification accuracy and validation but do not discuss calibration as a requirement for valid prevalence estimates (Grimmer et al. 2022); multicalibration is entirely absent from these discussions.

Applying multicalibration to LLM-based measurement. A natural question is how to define the subgroup collection $\mathcal{G}$ when the measurement device is an LLM. In many applications, multicalibration does not require access to the model’s internals. When LLMs are used to classify documents, survey responses, or clinical notes, the inputs typically carry observable metadata—document source, author demographics, text length, topic, or language—that can serve directly as segment features for multicalibration. These metadata features often correspond precisely to the dimensions along which population composition may shift.

For richer subgroup structure, multicalibration can also operate on LLM embedding representations. Detommaso et al. (2024) demonstrate two approaches for defining subgroups over prompt-completion pairs: clustering within the embedding space, and “self-annotation”—querying the LLM with yes/no questions

about properties of the input (e.g., “Does this question require specialized knowledge?”) to generate binary group indicators. Both strategies yield subgroups that are correlated with the probability of correct classification, enabling effective multicalibration of LLM confidence scores. These techniques provide a concrete path toward multicalibrated LLM-based measurement, though developing end-to-end pipelines for prevalence estimation remains an active area of work.

Several limitations warrant discussion. First, multicalibration requires specifying the collection of subgroups $\mathcal{G}$. The guarantee holds only for shifts along axes represented in $\mathcal{G}$; if the relevant dimension of shift is absent, bias can persist—as reflected in the modest OOD bias in our empirical results, where geographic shift introduces variation beyond the calibrated features. In practice, $\mathcal{G}$ should include all features plausibly associated with both the outcome and the shift. Second, multicalibration requires labeled calibration data of sufficient size to estimate subgroup-specific corrections. This is feasible for standard ML pipelines but more challenging for LLMs used in truly zero-shot settings, where obtaining calibration labels may require the manual annotation that LLMs were intended to avoid. Prediction-powered inference (Angelopoulos et al. 2023) offers a complementary framework for combining small labeled datasets with large-scale LLM predictions. Third, modern LLMs produce text rather than probability scores; calibratable scores can be obtained via token log-probabilities, confidence-elicitation prompts, or repeated sampling. Recent work on multicalibration for LLM confidence scoring (Detommaso et al. 2024) provides practical strategies for defining subgroups in this setting (see discussion above), but end-to-end multicalibration pipelines for LLM-based prevalence estimation remain an active area of development. Fourth, our framework assumes covariate shift ($P(Y|X)$ stable). Under concept drift, where the feature-outcome relationship itself changes, no purely statistical correction can substitute for new labeled data from the target distribution. Finally, our empirical application uses synthetic distribution shifts to evaluate bias. This design choice is deliberate: measuring bias requires knowing the true prevalence in the target population, which is generally unavailable under naturally occurring shifts. The synthetic shifts we construct are realistic in magnitude and correspond to substantively meaningful population differences.

These results connect two literatures that have developed in isolation. The quantification literature has focused on correcting aggregate error rates but has not engaged with subgroup-level calibration (González et al. 2017; Wu and Resnick 2024b). The multicalibration literature has focused on individual-level prediction quality and fairness but has not emphasized the implications for population-level inference (Hébert-Johnson et al. 2018; Kim et al. 2022). Our contribution is to show that subgroup-level calibration is the missing link that makes model-based prevalence estimation reliable under the distribution shifts that motivate its use.

Materials and Methods

Simulation. We generate $n = 10,000$ observations with a binary covariate $X \in \{0, 1\}$ and outcome $Y \sim \text{Bernoulli}(P(Y|X))$, where $P(Y = 1|X = 1) = 0.85$ and $P(Y = 1|X = 0) = 0.15$. The classifier produces deterministic scores $\hat{p}(X = 0) = 0.135$ and $\hat{p}(X = 1) = 0.935$, representing 10% multiplicative bias within each stratum. For each of $B = 50$ iterations, we generate fresh calibration data from $P(X = 0) = 0.5$, estimate all method-specific parameters, then evaluate bias and mean squared error on 20 test distributions with $P(X = 0)$ ranging from 0.01 to 0.99. The five methods compared—uncalibrated averaging, Classify & Count with a prevalence-matched threshold, Rogan-Gladen adjustment, global multiplicative calibration, and multicalibration with stratum-specific additive corrections—are detailed with full mathematical definitions in the SI Appendix.

Empirical application. We use the ACS Public Use Microdata Sample via the *folktables* package, with the ACSEmployment prediction task (binary: employed vs. not employed) and 16 sociodemographic features. Training data comprises eight states (TX, MI, PA, OH, IL, GA, NC, VA) across 2016–2018. The base model is logistic regression with standard scaling. Post-hoc calibration uses isotonic regression (global) and MCGrad (multicalibration with categorical and numerical segment features) on a held-out calibration set ($n \approx 644,000$). Synthetic age-shifted populations are created by importance-weighted resampling of the test set ($n \approx 920,000$), with exponential weights favoring young ages, old ages, or both extremes. RMSE is computed via bootstrap resampling (200 iterations per scenario). Six additional states (CA, NY, FL, WA, AZ, CO) serve as OOD evaluation data.

Software. MCGrad is available at github.com/facebookincubator/MCGrad. Simulation and analysis code are available at [repository URL].

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